

Acid/Base Balance and Fluid Homeostasis

Today's Materials

- Activity 1: Fluid Homeostasis
 - De-shelled raw chicken eggs (4/group)
 - 250mL beakers (4/group)
 - Salt solutions of 4% and 20% in water
 - DI water
 - Unknown salt solution
 - Scales
 - Weigh boats
 - Graph paper
 - Gloves
- Activity 2: Acid/Base Balance of Cardiovascular and Respiratory Systems
 - Small Test tubes
 - Test tube racks
 - Narrow range pH paper
 - Deionized water and dropper
 - Animal (sheep) plasma and dropper
 - 0.01 M HCl dropper bottles
 - 250 mL beakers
 - 0.05 M NaOH dropper bottles
 - Phenol Red
 - Straws

Activity 1: Fluid Homeostasis

Adapted from J. Kurushima and Gavilan College General Biology Lab Manuals.

More than half of your body mass is made up of fluids (on average, this is over 90 pounds of water!). For the most part, body fluids are kept in two “compartments”: (1) intracellular fluid (ICF) or cytosol, and (2) extracellular fluid (ECF). In addition, there are many **fascial compartments** throughout the body. A fascial compartment is a region of muscle isolated by fascia. If a cell or tissue is to perform its functions, it must maintain a steady state in the midst of an ever-changing environment. In this activity you will observe the osmotic movement of water between the extracellular fluid and the intracellular fluid of a chicken egg (a very large cell).

We use the terms **hypotonic**, **hypertonic** and **isotonic** in referring to the relative concentrations of solute particles of different solutions.

1. Define the three tonicity terms (hypotonic, hypertonic and isotonic) (you may need to reference your notes from Bio40A). Be sure to reference the concentrations of both the solute and the solvent in your definitions.
2. With your group, use weigh boats to obtain 4 chicken eggs from which the shell has been dissolved away. The remaining membrane (the shell membrane) is differentially permeable and contains the

cytosol of the cell. We will assume that each egg contains approximately the same concentration of solutes.

3. Weigh each egg separately to the nearest 0.1g and record the weights in Table 1 at time "0".
4. Place eggs 1, 2, and 3 into separate beakers containing solutions of distilled water, 4% salt, and 20% salt. Place egg 4 into a beaker containing the "unknown" solution.
5. At 15 minute intervals, that is after 15, 30, 45, 60, and 75 minutes, remove the eggs from the beakers; carefully wipe off all excess water; and again, weigh each egg separately. Record the weight in Table 1.
6. Calculate the change in weight of the eggs over time. (This allows us to compensate for the differences in starting weight between eggs.) Record the change in weight over time in Table 2.
7. Plot the changes in weight of each of the eggs against time on a piece of graph paper.
8. Answer the following questions in your notebook:
 1. Compare the solutions to the eggs:
 - a. Which solutions would you say were hypotonic to that of the eggs?
 - b. Which of them were hypertonic?
 - c. Isotonic?
 2. What would you expect to happen if an egg was put into a fifth beaker containing a 40% salt solution?
 3. In this experiment, the egg represented a body cell, while the beaker fluid modeled the extracellular fluids in the body.
 - a. What would happen to your blood cells if you drank a large volume of water, which diluted your blood plasma?
 4. We can also apply these concepts to the fascial compartments found in muscles. Compartment syndrome occurs when body fluids cannot move between the compartments of the body, and fluid becomes trapped between the fascia and the musculature underneath. This fluid will exert enough pressure to eventually damage the underlying tissues. Compartment syndrome may result from a traumatic injury (such as a severe hematoma, fracture, or surgery), or in some cases tightly wrapping or binding a limb (such as in a cast, tourniquet, or in one case, wearing tight "skinny jeans").
 - a. Perform an internet search to find the common signs and symptoms of compartment syndrome.
 - b. How is compartment syndrome treated?
 - c. Based on your internet research, describe a few different ways that abnormalities in osmosis contribute to compartment syndrome.
 5. Return the eggs to the designated container. Use the funnels to return salt solutions to their appropriate original jugs. Make sure to place the correct solutions back into the correct jugs!

Table I: Weight of Eggs (g) vs Time (minutes)

Time (Min.)	0% salt	4% salt	20% salt	Unknown
0				
15				
30				
45				
60				

Table II: Weight Change of Eggs (g) vs Time (minutes)

Time (Min.)	0% salt	4% salt	20% salt	Unknown
0	0	0	0	0
15				
30				
45				
60				

Activity 2: Acid/Base Balance of the Cardiovascular and Respiratory Systems

Adapted from "Role of Respiratory in Acid-Base Balance in Blood"

As cells undergo cellular respiration they produce carbon dioxide (CO_2) as a byproduct. This carbon dioxide is released into the blood where it combines with water to create carbonic acid (H_2CO_3) which will lower the pH. Blood pH must be maintained around pH 7.4 for the body cells to function. The carbonic acid-bicarbonate system acts as a buffer in the blood to maintain pH. When the blood becomes too acidic ($< \text{pH } 7.4$) bicarbonate ions will bind to excess H^+ , if the pH is too alkaline ($> \text{pH } 7.4$) carbonic acid will release H^+ into the blood plasma.

Part 1: The role of the Cardiovascular System in Acid/Base Balance

Here we will compare the ability of blood plasma to buffer changes in pH to that of distilled water.

- Obtain 2 test tubes.
- Add 2 mL of animal blood plasma into the first test tube.
 - Use a pH test strip to measure the pH of the blood plasma. _____
 - Add 2 drops of 0.01 M HCl (hydrochloric acid) into the test tube. Use a fresh pH test strip to measure the pH of the blood plasma + 2 drops HCl. _____
- Add 2 mL of water to the second test tube.
 - Use a pH test strip to measure the pH of the water. _____
 - Add 2 drops of 0.01 M HCl (hydrochloric acid) into the test tube. Use a fresh pH test strip to measure the pH of the water + 2 drops HCl. _____
- Answer the following questions:
 - Is the plasma a good buffer?
 - What component of the blood plasma carbonic acid-bicarbonate buffer system was acting to counteract a change in pH when HCl was added?
- Pour the plasma and water in the appropriate waste bottles.

Part 2: The role of the Respiratory System in Acid/Base Balance

Now we will observe how exhaling CO_2 helps the body to maintain its acid/base balance.

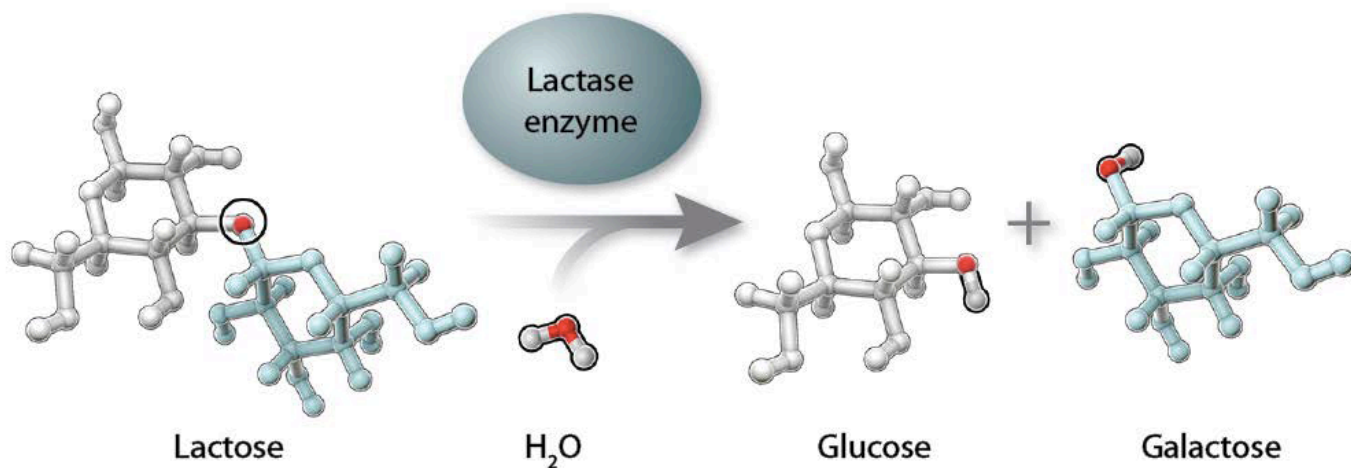
- Fill a beaker with 100 mL of distilled water

2. Add 0.5 mL of 0.05 M NaOH and 2 drops of phenol red. Swirl.
 - a. Phenol red is a pH indicator, it is magenta in alkaline solutions, red in neutral solutions, and yellow in acidic solutions.
 - b. Is the solution currently alkaline, neutral or acidic?
3. Use a straw to gently blow bubbles into the solution until it changes color.
 - a. Is the solution now alkaline, neutral or acidic?
 - b. As you exhale into the beaker, what chemical reaction is taking place?
4. Discard the straw in the garbage, and pour the phenol red solution in the appropriate waste bottle.

Activity 3: Acid/Base Balance in the Digestive System

Adapted from “Got Lactase? Blood glucose data analysis” HHMI Biointeractive.org

“Mother’s milk” is packed with the proteins, fats, and carbohydrates that support the growth, development, and survival of baby animals. The sugar lactose is the main carbohydrate in milk. Lactose can be cleaved into two simple sugars, glucose and galactose, by lactase, an enzyme produced in the small intestine. The two smaller sugars are readily absorbed through the intestinal wall into the bloodstream for delivery to the cells of the body, where they are used for energy.



After infant mammals are weaned from their mother’s milk, lactase production shuts down, presumably because it is no longer needed. This condition is called lactase nonpersistence—meaning that production of the lactase enzyme does not *persist* into adulthood. The general condition for mammals is not to consume milk after weaning and to be lactase nonpersistent. Some populations of humans are unusual in that adults continue to consume milk from other mammals, such as cows.

If a person who is lactase nonpersistent drinks milk, undigested lactose passes from the small intestine to the large intestine, where it is fermented by bacteria. Fermentation produces various gases in the large intestine, which can cause abdominal pain, bloating, flatulence, and diarrhea—all symptoms of lactose intolerance. Worldwide, most adults are lactose intolerant, although some people may not know it because their symptoms are mild. Only a minority of human adults (about 35% of the global human population) continues to produce lactase into adulthood and can drink milk without any problems. These individuals are said to be lactase persistent or lactose tolerant.

Part 1: Blood Glucose Test to Determine Lactose Tolerance

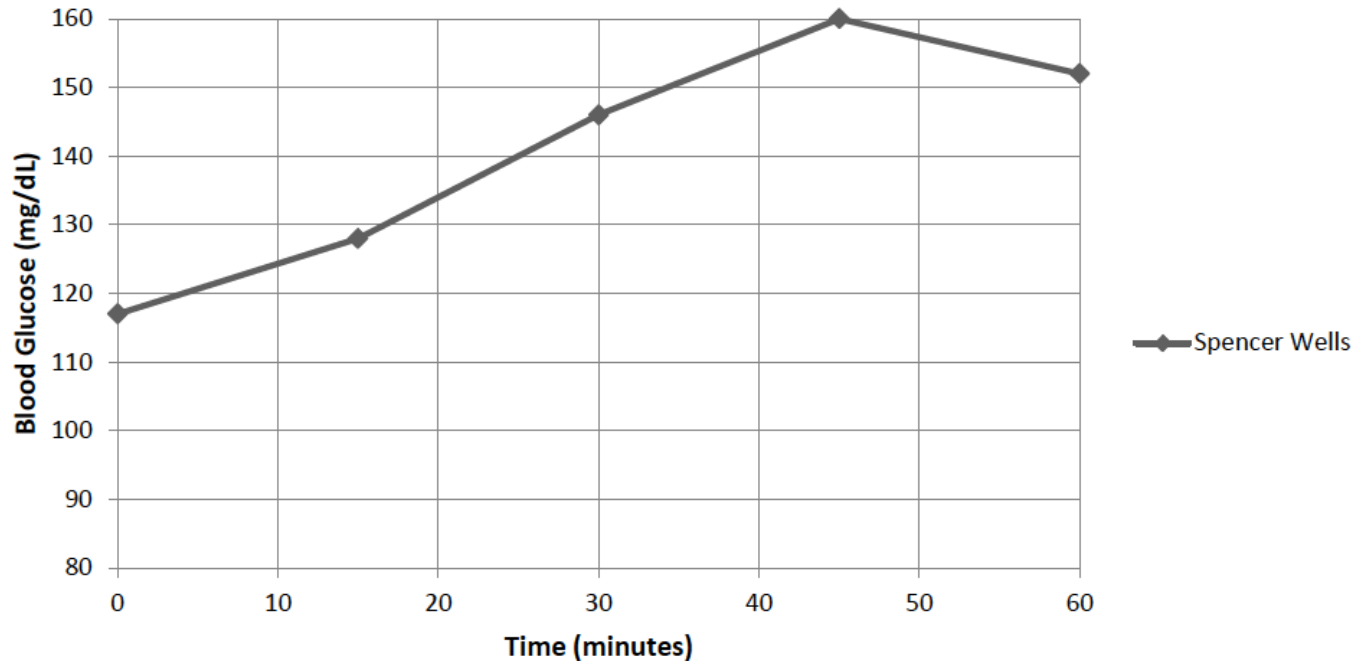
There are several ways to test whether someone is lactase persistent. One method, is to have the patient fast overnight, and then drink one liter of milk. If the patient is lactose tolerant, the lactose will be cleaved by lactase into glucose and galactose, which will then be absorbed into the blood stream. You can then measure the blood glucose of the patient. (For a short video demonstrating this process, watch <https://www.hhmi.org/biointeractive/making-fittest-got-lactase-co-evolution-genes-and-culture> up to the 4:30 mark.)

1. Examine the data in the table below. It shows the blood glucose levels of six individuals tested in Dr. Sarah Tishkoff's laboratory. After baseline (i.e., time 0 minutes) blood glucose levels were measured and recorded, each person drank a liter of milk. Blood glucose levels were again measured at 15, 30, 45, and 60 minutes after drinking the milk. Glucose levels were measured using glucose strips and a glucose reader similar to the one Dr. Wells used in the film.

	Blood Glucose (mg/dL)				
Individual	0 minutes	15 minutes	30 minutes	45 minutes	60 minutes
Spencer Wells	117	128	146	160	152
Peter	97	111	135	154	143
Rachel	96	99	105	101	98
Katherine	95	97	99	101	102
Sarah	108	116	129	141	139
Michael	94	109	128	143	140
Arthur	97	96	94	83	88

2. Plot the results from the six individuals in the graph below. The graph already includes Dr. Wells' blood glucose results. Make sure to include a legend for your graph.

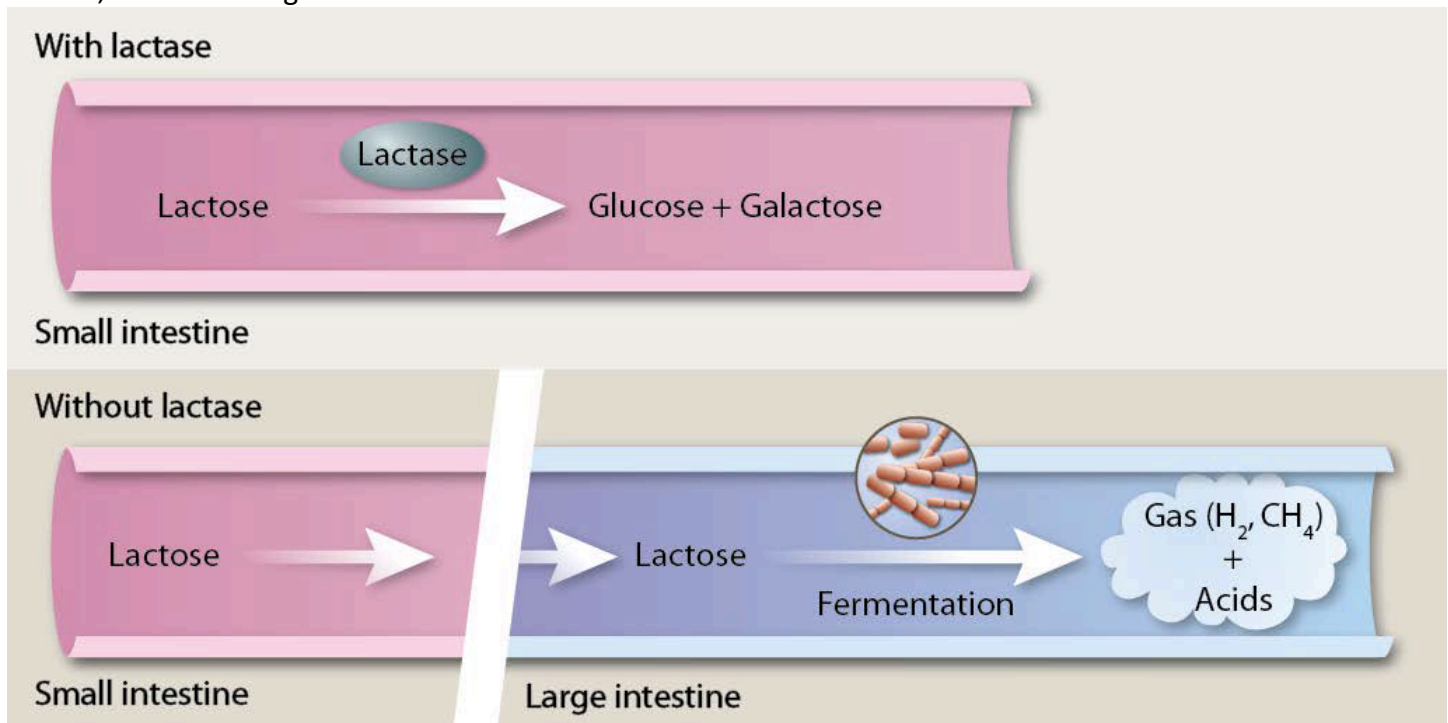
Lactase Persistence Blood Glucose Test Results



3. Answer the following questions in your notebook.
- Why is measuring blood glucose levels an indicator of someone's lactase activity?
 - Based on your graph, which individuals are lactose persistent? Which are lactose intolerant? Explain your rationale.
 - A person taking a blood glucose test is usually told to fast prior to the test. Why do you think that might be necessary?

Part 2: Hydrogen Breath Test to Determine Lactose Tolerance

Another common test used to determine whether a person is lactose persistent is the hydrogen breath test. This test measures the amount of hydrogen in a person's breath. As you read above, undigested lactose is fermented by bacteria in the large intestine, producing several gases, including hydrogen. These gases exit the body through the anus; they can also be absorbed into the blood, circulated to the lungs, and eliminated through the breath.



- Review the data in the following table. It shows the hydrogen breath levels of four individuals tested for lactase persistence. Time 0 represents the time before drinking milk and the other times are times after drinking milk.

	Hydrogen Breath Levels (ppm)				
Individuals	0 minutes	30 minutes	60 minutes	90 minutes	120 minutes
Lisa	5	6	9	8	5
Dan	4	9	8	29	35
Cindy	6	8	10	31	32
Brian	4	7	6	9	6

- Graph the data for the individuals in the table above, including a title, labels for the x- and y-axes, and a legend.
- Answer the following questions in your notebook.
 - Which individuals appear to be lactase persistent? Use data to support your answer.
 - Which individuals appear to be lactase nonpersistent? Again, use data to support your answer.
 - As hydrogen gas from the intestines enters the blood stream, will this affect the blood plasma? (Remember, hydrogen gas is H_2). Do you expect this to alter the blood pH? Why or why not?
 - Can you think of another test you might do to determine a person's lactase status?